

STAT 1 Course schedule (preliminary)

Main subject	#	Topics	Literature [MR] = [Montgomery & Runger]	Problems dataset available on BB [SP] = Summarizing Problem [BB]
Basics	1	Basic definitions and theory of probability	[MR]: 1 (skim) 2	[MR]: 2.1.13, 2.1.22, 2.2.7, 2.2.10, 2.3.8, 2.4.2, 2.5.2, 2.5.7, 2.6.5, 2.7.4, 2.7.11-12, 2.8.1
Statistical modeling	2	Discrete random variables and probability distributions	[MR]: 3 (skip 3.7)	[MR]: 3.1.10, 3.1.11, 3.2.8, 3.3.1, 3.3.7, 3.4.1, 3.5.2, 3.5.4, 3.5.13, 3.6.1, 3.8.2
	3	Continuous random variables and probability distributions	[MR]: 4.1-4.7	[MR]: 4.1.2, 4.2.4, 4.3.1, 4.4.1, 4.5.1, 4.5.4, 4.5.13, 4.6.1, 4.7.13
	4	Joint probability distributions	[MR]: 5.1-5.7	[MR]: 5.1.2, 5.1.9, 5.4.1, 5.4.7, 5.6.1, 5.6.9, 5.7.3
Estimation of model parameters	5	Descriptive statistics Sampling distributions Point estimation	[MR]: 6+7 (skip 7.4.1 and 7.4.3)	[MR]: 6.1.1-5, 6.1.9, 6.1.12, 6.3.3, 6.4.1, 6.7.2, 7.2.4, 7.3.2, 7.4.1 → [SP]: 1,2,3,4
	6	Interval estimation (confidence intervals)	[MR]: 8 (skip 8.6)	[MR]: 8.1.3, 8.1.4, 8.1.12, 8.2.1, 8.2.9, 8.2.12, 8.3.3, 8.4.1 → [SP]: 5
Statistical hypothesis testing	7	Single-sample tests	[MR]: 9.1-9.6	[MR]: 9.2.2, 9.2.3, 9.2.4, 9.2.6a, 9.2.7, 9.3.1, 9.3.7, 9.4.1, 9.4.3, 9.5.2 → [SP]: 6
	8	Two-sample tests	[MR]: 10 (skip 10.3)	[MR]: 10.1.1a+b, 10.2.2a+b, 10.2.6, 10.2.11a+b+c, 10.4.1, 10.4.3 → [SP]: 7,8,9
Correlation, regression and prediction	9	Regression models: simple linear, non-linear, multiple linear and polynomial	[MR]: 11.1, 11.2, 11.7, 11.9 (skim 12.1)	[MR]: 11.2.1, 11.2.3, 11.2.6, 11.2.10, 11.7.3, 11.7.5, 11.9.1, 11.9.2 → [SP]: 10,11
	10	Inference for simple linear regression	[MR]: 11.3-11.6	[MR]: 11.4.1, 11.4.2, 11.4.5, 11.4.8, 11.6.1, 11.6.3, 11.7.2 → [SP]: 12,13,14
Comparison of several samples	11	ANOVA 1: One-factor analysis of variances	[MR]: 13.1-13.2	[MR]: 13.2.1, 13.2.2, 13.2.3, 13.2.8 → [SP]: 15,16
	12	ANOVA 2: Two-factor analysis of variances	[MR]: 14.1-14.4 (skim 13.4)	[MR]: 14.3.1, 14.3.2, 14.3.3a-d, 14.3.7

Information about STAT 1

Course literature

- Montgomery & Runger:
Applied Statistics and Probability for Engineers
7'th Edition, EMEA Edition, 2019, Wiley
ISBN 978-1-119-58559-6
- Various material available on itslearning

Course SW

MATLAB is mandatory for solving problems in class and at the exam
Available for download from itslearning

Course activities

Theory and problems including usage of MATLAB

Course assessment

Written multiple choice exam, 3 hours, 25 problems, 7-level grade, internal censor

Teacher

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